Modelling and Analyis of Microstructures of Electronic Markets

MA 499 Project II

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Outline

1. Introduction

2. Market Microstructure Theories

3. Mathematical Modelling and Numerical Simulations



Introduction

Evolution to Electronic Limit Order Books

- Early financial markets were mainly of two types:
  - **Open Outcry Markets:** Physical trading in a shared space
  - **Quote Driven Markets:** Dealers posted bid and ask quotes
- These systems had high transaction costs, low transparency, and execution delays
- Exchanges like **CBOT, CME, LIFFE** used Open Outcry, while **NYSE, NASDAQ, LSE** were Quote Driven Markets
- To improve speed and transparency, markets shifted to **order driven electronic exchanges**
- **Toronto Stock Exchange** introduced CATS in 1977, and **Paris Bourse** became the first fully electronic LOB-based equity market
- This transition reduced latency and enabled algorithmic and high-frequency trading



Limit Order Books and Order Book Dynamics

- A **Limit Order Book (LOB)** stores all buy/sell orders in an electronic market
- Each order contains information such as:
  - Buy/Sell side, Price, Quantity, and Timestamp
- The **Ask Price** is the lowest priced sell limit order
- The **Bid Price** is the highest priced buy limit order
- The **Bid-Ask Spread** is the difference between the Ask and Bid prices
- Orders are executed using **price-time priority (FIFO)**
- Three main types of orders:
  - **Limit Orders:** Specify price and quantity for buying or selling
  - **Market Orders:** Execute immediately at the best available quote
  - **Cancellation Orders:** Remove an existing limit order
- The evolution of the LOB reflects market movements and supports price discovery, trading strategies, and market regulation

The Need to Model Limit Order Books (LOBs)

- **LOBs** are the core mechanism through which buyers and sellers interact
- They capture the price formation dynamics at the most granular level
- LOB data is widely used in **algorithmic & high-frequency trading** for *designing*, *calibrating*, and *fine-tuning* automated trading strategies
- Modelling LOBs helps study important market phenomena such as:
  - Volatility clustering
  - Correlated order flow
  - Price impact and market impact
  - Liquidity dynamics
- Current techniques of simulating LOBs include Point-Processes models, Agent based models, Deep Learning based models and Stochastic Differential Equations based models



Importance of LOB Simulation

- Despite challenges, LOB simulation is essential for **algorithmic trading research** and strategy development
- A major use case is **backtesting trading strategies**, where simulators provide richer datasets beyond historical market realizations
- Relying only on one historical price path can lead to overfitting and poor true out-of-sample performance
- Synthetic data generated through simulators helps reduce overfitting and improves strategy robustness
- The simulator must reproduce important **stylized facts** of real markets to remain realistic and avoid bias
- Best practice combines simulated data with real market data for reliable training and evaluation



Market Microstructure Theories



Market Microstructure and Market Participants

- **Market Microstructure** studies how trading rules and market structure affect price formation, liquidity, and price discovery
- In electronic markets driven by Limit Order Books (LOBs), participants are broadly classified into:
  - **Market Makers (MMs):** Provide liquidity by posting bid and ask quotes and earn the spread
  - **Informed Traders:** Trade using private information before it is reflected in prices
  - **Fundamental / Noise Traders:** Trade for hedging, consumption, or portfolio rebalancing and are less sensitive to short-term prices
- The interaction between these participants determines liquidity, spread formation, and short-term price dynamics
- These concepts form the foundation for algorithmic and high-frequency trading strategies



Foundational Models of Market Microstructure

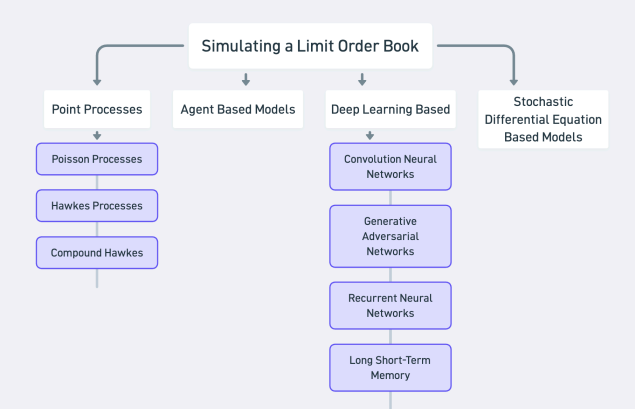
- **Grossman-Miller Model (1988)**
  - Explains inventory risk faced by market makers
  - Spread compensates for the risk of holding temporary inventory
  - Introduces the idea of temporary price impact
- **Kyle Model (1985)**
  - Studies informed trading and price discovery
  - Introduces Kyle's Lambda (λ) as permanent price impact
  - Price adjusts according to the information content of order flow
- **Glosten-Milgrom Model (1985)**
  - Explains spread as protection against adverse selection
  - Market makers adjust bid-ask spread based on informed trading risk



Mathematical Modelling and Numerical Simulations

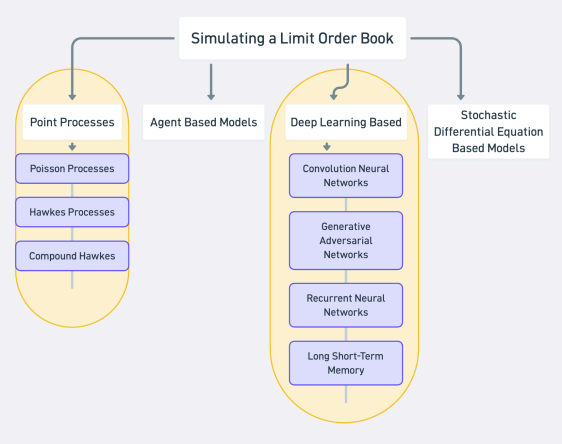


Order Book Modelling



Different Paradigms of Order Book Modelling

Order Book Modelling



Our focus in this project

Mathematical Modelling and Numerical Simulations

Point Processes based Models

Poissonian Order Flows

- Model parameters: K, λ_M, $\{\lambda_{L,i}\}$, $\{\lambda_{C,i}\}$, a_∞, b_∞
- Simulation inputs: N (number of events), X_{init} (initial book state)
- Update cancellation rates using current queue depths
- Sample waiting time τ from an exponential distribution
- Select event type based on order-flow intensities
- Sample event volume and price level if applicable
- Update order book state and advance time
- Enforce boundary depths a_∞ and b_∞

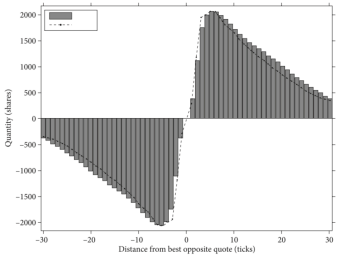


Poissonian Order Flows - Parameter Estimation

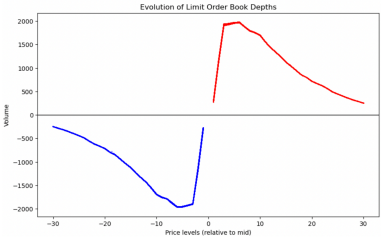
- T: Length of the daily observation window
- N_T^M: Total number of market orders observed during T
- Market order intensity: $\lambda_M = \frac{N_T^M}{2T}$ (average arrival rate of market orders)
- $N_{i,T}^L$: Number of limit orders submitted i ticks from the best opposite quote
- Limit order intensity at level i: $\lambda_{L,i} = \frac{N_{i,T}^L}{2T}$ (arrival rate of limit orders at level i)
- $N_{i,T}^C$: Number of cancellations at level i
- $\langle X_i \rangle$: Average resting depth at level i
- Cancellation intensity: $\lambda_{C,i} = \frac{N_{i,T}^C}{2T\langle X_i \rangle}$ (cancellation rate normalized by available volume)
- Daily estimates averaged over number of trading days
- Order sizes modeled using log-normal distributions

Poissonian Order Flows - Results

We reconstruct the depth profile reported for SCHN.PA(Abergel 2016) using the given parameter values.



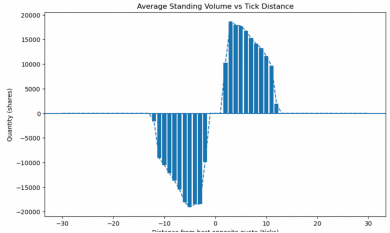
Empirical Average Depth Profile SCHN.PA



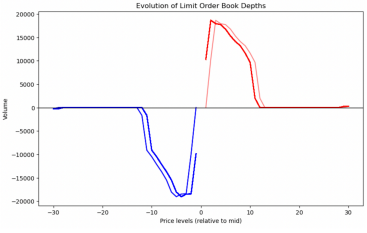
Simulated Depth Profile SCHN.PA

Poissonian Order Flows - Results

We used the algorithm in order to reproduce the depth profile of the LOB for data external to the text to fulfill an ablation study.

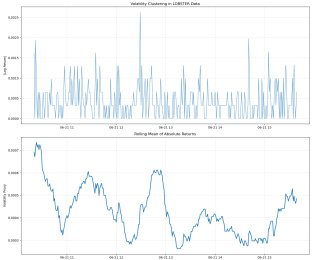


Empirical Depth Profile MSFT computed from LOBSTER L10 Data

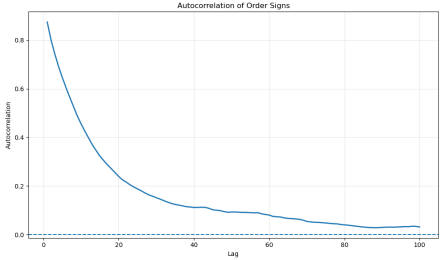


Simulated Depth Profile MSFT computed from estimated parameters

Limitations of Poisson Processes



Volatility Clustering Observed in MSFT Data



Autocorrelation observed in MSFT Data

Mathematical Modelling and Numerical Simulations

Hawkes Processes



Motivation for Hawkes Processes

- Poisson models assume **independent arrivals**, which is little unrealistic
- Real market data shows:
  - **Clustering of events**
  - **Self-excitation in order flow**
  - **Volatility clustering**
- To capture these effects, we use **Hawkes Processes**
- Key idea:
  - *Past events increase the probability of future events*
- Widely used in modelling:
  - Market orders, limit orders, cancellations
  - High-frequency trading dynamics



The Hawkes Process

- A **Hawkes process** is a self-exciting point process
- Intensity depends on history:

$$\lambda(t) = \mu + \sum_{t_i < t} \phi(t - t_i)$$

- Components:
  - μ: baseline intensity
  - $\phi(\cdot)$: excitation kernel
- Common choice:

$$\phi(t) = \alpha e^{-\beta t}$$

- Interpretation:
  - Each event increases future intensity
  - Effect decays exponentially over time



Multivariate Hawkes Processes in LOB

- Real markets involve multiple event types:
  - Market orders
  - Limit orders
  - Cancellations
- Use **multivariate Hawkes processes**
- Intensity for type i:

$$\lambda_i(t) = \mu_i + \sum_j \sum_{t_k^j < t} \phi_{ij}(t - t_k^j)$$

- Captures:
  - Self-excitation ($i = j$)
  - Cross-excitation ($i \neq j$)
- Models interaction between different order types

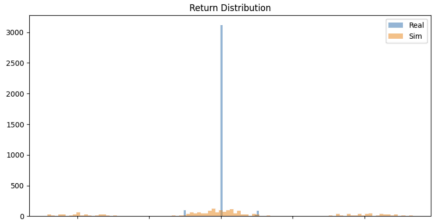


Simulation of Hawkes-based LOB

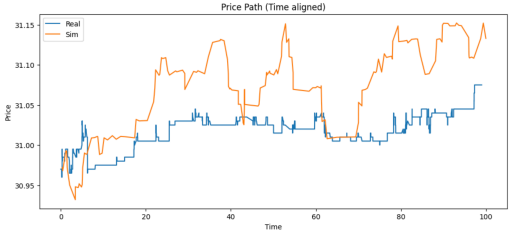
- Simulation is event-driven (similar to Poisson case)
- Key steps:
  - Compute intensities $\lambda_i(t)$
  - Sample next event time
  - Select event type using intensities
  - Update intensities dynamically
- Requires:
  - Kernel parameters (α, β)
  - Baseline intensities μ_i
- More complex than Poisson:
  - State-dependent and history-dependent

Hawkes Processes - Results

- Hawkes models successfully reproduce: **Volatility Clustering, Correlated Order Flows, Realistic Inter-Arrival Times**
- Matches empirical observations:
  - Bursts of high activity followed by calm periods



Returns generated via Hawkes Processes



Simulated versus actual Price Path

Hawkes Processes - Limitations

- Calibration is computationally expensive
- Model complexity increases with:
  - Number of event types
  - Kernel parameters
- Requires large datasets for stable estimation
- Despite challenges:
  - Provides much more realistic dynamics than Poisson models

Mathematical Modelling and Numerical Simulations

Compound Hawkes Processes



Compound Hawkes Process (CHP)

- Extends Hawkes process by including **random marks**
- Each event has:
  - Arrival time (from Hawkes process)
  - **Associated size or volume**
- Model:

$$X(t) = \sum_{i:t_i \leq t} V_i$$

- V_i represents:
  - Order size
  - Price impact

Why Compound Hawkes?

- Standard Hawkes models only **event timing**
- Real markets require modelling:
  - **Order sizes**
  - **Impact of trades**
- CHP captures:
  - Clustering in arrivals
  - Heavy-tailed order size distributions
- Useful for:
  - Price dynamics modelling
  - Market impact analysis



Compound Hawkes - Results

- Better captures:
  - **Heavy-tailed returns**
  - **Price jumps**
  - **Order size clustering**
- Matches empirical stylized facts:
  - Large orders tend to cluster
  - Market impact persists over time
- Improves over standard Hawkes:
  - Adds realism in volume modelling

Mathematical Modelling and Numerical Simulations

Deep Learning based Methods



AI Architectures for Limit Order Book Simulation

- Recent advances in deep learning have significantly improved financial market simulation.
- Architectures such as:
  - Convolutional Neural Networks (CNNs)
  - Recurrent Neural Networks (RNNs)
  - Long Short-Term Memory Networks (LSTMs)
  - Generative Adversarial Networks (GANs)

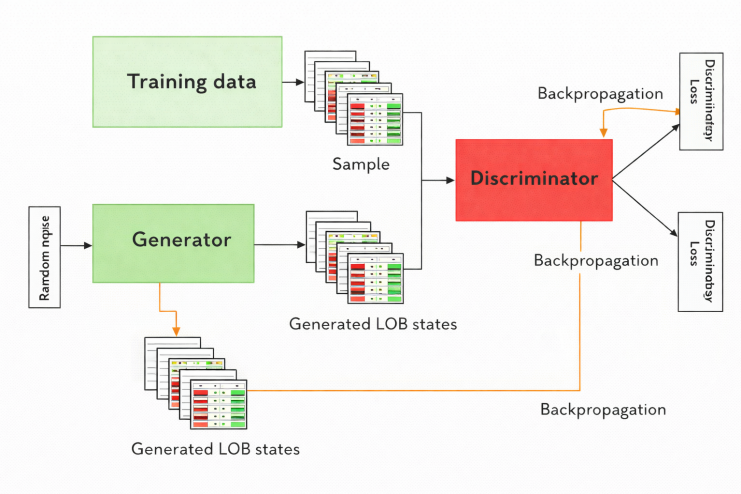
  are increasingly used to model complex Limit Order Book (LOB) dynamics.
- These models capture:
  - Nonlinear dependencies
  - Temporal evolution of order flow
  - Stylized facts such as volatility clustering and market impact
- Goal: Learn realistic future LOB states conditioned on current market state.

Mathematical Modelling and Numerical Simulations

Generative Adversarial Networks



GAN Architecture for Limit Order Book Simulation



GAN Architecture for Limit Order Book Simulation

- **Generator** takes random noise z and current LOB state X_t as input
- Produces synthetic future LOB snapshot $\hat{X}_{t+\Delta t}$
- **Critic (Discriminator)** compares:
  - Real future state $X_{t+\Delta t}$
  - Generated future state $\hat{X}_{t+\Delta t}$
- Wasserstein loss + Gradient Penalty ensures stable training
- Objective: learn realistic conditional LOB transitions

Using GANs to Model the Limit Order Book

- LOB represented as a **centered snapshot** around mid-price:
  - Buy queues: negative values
  - Sell queues: positive values
- Model learns:

$$P(X_{t+\Delta t} \mid X_t)$$

  i.e., future LOB state conditioned on current state.
- Markovian setting:
  - Condition = current LOB snapshot only
- Symmetry enforcement:
  - Random flipping of LOB with 50% probability
  - Prevents artificial directional price drift

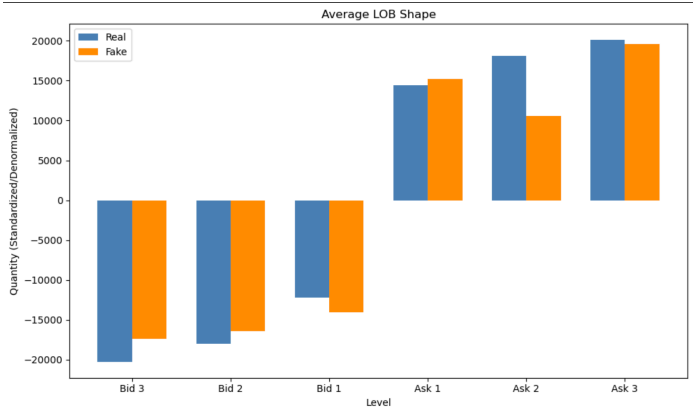
LOB State Representation

The limit order book is represented as a **centered LOB snapshot** around the mid-price:

$$X_t = \left(Q_{p(t)-(k-1)\delta}(t), \ldots, Q_{p(t)-\delta}(t), Q_{p(t)+\delta}(t), \ldots, Q_{p(t)+(k-1)\delta}(t) \right)$$

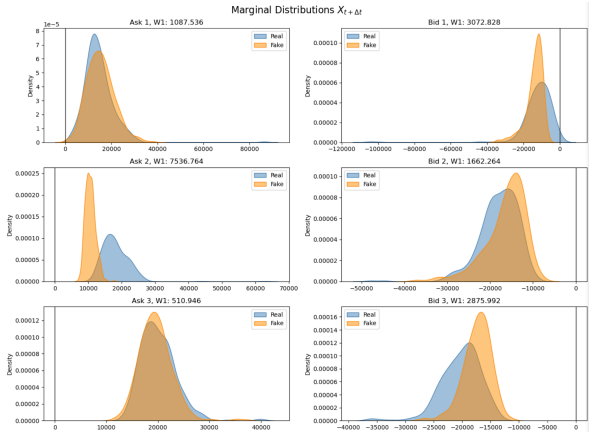
- $Q_p(t)$ denotes the queue size at price level p at time t
- **Bid-side queues** are represented using negative values
- **Ask-side queues** are represented using positive values
- The snapshot is centered around the mid-price and captures local LOB dynamics
- In our implementation, we use $k = 3$
  - 3 bid, 3 ask queues, giving a 6-level LOB representation

Average LOB Shape Results



Average Shape of LOB Simulations

LOB Marginal Density Results



Marginal Densities of LOB Simulations

Mathematical Modelling and Numerical Simulations

CNN-LSTMs



Order Flow Representation and Stationarity

- Raw Limit Order Book (LOB) states form a highly complex and non-stationary process
- To improve learning, the raw LOB is transformed into:
  - **Order Flow (OF)** — 20-dimensional vector capturing queue changes between consecutive states
  - **Order Flow Imbalance (OFI)** — 10-dimensional vector obtained by subtracting ask flow from bid flow
- These transformations help create a more stationary process
- Benefits:
  - Easier learning of long-term dependencies by LSTM
  - CNN layers can combine bid and ask flows asymmetrically
  - OF generally performs better than OFI for forecasting

CNN-LSTM Architecture for Price Movement Prediction

- **CNN Layers**
  - Reduce dimensionality of multivariate LOB input
  - Aggregate information across price levels and order sides
- **Inception Module**
  - Captures market behavior across multiple timescales
  - Acts like a learned moving average indicator
- **LSTM Layer**
  - Learns temporal dependencies from compressed sequences
- **Softmax Output Layer**
  - Predicts probabilities of future price movements
- **MC Dropout**
  - Reduces overfitting
  - Captures epistemic uncertainty in predictions

Future Scope of the Project

- **Hybrid Modelling Frameworks**
  - Combine explainable point-process models (Poisson/Hawkes) with deep learning models (GANs, CNN-LSTMs), such as Neural Hawkes Processes.
  - Improve both interpretability and predictive performance
- **Improved Calibration and Real-Time Adaptation**
  - Dynamic parameter estimation using high-frequency live market data
  - Adaptive models for regime shifts and intraday structural changes
- **LLM-Based Order Flow Simulation**
  - Tokenizing order book messages for autoregressive generation
  - Exploring transformer-based architectures for realistic LOB simulation
- **Execution and Market Making Applications**
  - Optimal execution strategies
  - Reinforcement learning for market making and inventory control

Concluding Remarks

- Limit Order Books form the core microstructure of modern electronic markets
- Point-process models such as Poisson and Hawkes processes provide:
  - Strong mathematical interpretability
  - Explicit modelling of order arrivals and market impact
- Deep learning models such as GANs and CNN-LSTMs provide:
  - Strong predictive performance
  - Ability to capture complex nonlinear dependencies
- The key trade-off remains:

Explainability vs. Complexity

- Future research lies in bridging this gap through robust, interpretable, and scalable hybrid models

Thank You

Questions?



Key References

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